Fiscal Year:	FY 2023	**Task Last Updated:**	FY 08/22/2023
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Can Polyamines Mitigate Plant Stress Response under Microgravity Conditions?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	53706-1580	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	09/16/2019	**End Date:**	09/15/2023
No. of Post Docs:		**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/15/2023 per NSSC information (Ed., 7/26/22).		
Key Personnel Changes/Previous PI:	July 2020 report: Dr. Shih-Heng Su is an Assistant Scientist in the Masson lab who has been contributing to this project since its inception.		
COI Name (Institution):	Su, Shih-Heng Ph.D. (University of Wisconsin-Madison)		
Grant/Contract No.:	80NSSC19K1483		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space exploration requires the design and implementation of bioregenerative life-support systems capable of maintaining breathable air, cleaning water, and producing food and fiber during space missions. Plants are expected to play key roles in these life-support systems. Therefore, it is important to understand the effects the microgravity environment has on plants to allow the design and development of better life-support systems for space exploration. Multiple studies have demonstrated that plants grown under microgravity display signs of stress, including altered morphology and expression profiles expected for stress response. Therefore, the development of mitigation strategies aimed at alleviating such stress will be important. Previous studies by multiple research teams including ours have demonstrated a role for polyamines in stress mitigation in plants. Polyamines are a group of compounds whose abundance increases in plants under stress. These signaling molecules have also been shown to contribute to stress mitigation in ground-based experiments. Therefore, we hypothesize that polyamines might also contribute to stress mitigation under microgravity, and propose to test this hypothesis by investigating the growth and expression profiles of Arabidopsis thaliana seedlings engineered to modify the pool of putrescine (one of these polyamines) accumulating in their tissues. Specific objectives will include: 1) Comparing the growth of Arabidopsis thaliana seedlings with genetic alterations affecting the pool of putrescine to that of wild type, within the VEGGIE hardware, both under microgravity on the International Space Station (ISS) and on the ground (1-g control), and 2) Use RNAseq approaches to investigate the effect of microgravity relative to 1-g ground control on the transcriptomes of wild type and mutant seedlings with alterations in their putrescine pool. These investigations should lead to a better understanding of the roles played by polyamines in stress mitigation under microgravity. It may identify novel genetic engineering methods that improve plant growth under microgravity, thereby allowing development of optimized bioregenerative life-support systems for long-term space exploration missions. This project may also have important applications in agriculture for crop production on marginal lands. Finally, our experiments will yield large sets of transcriptomic data from previously uncharacterized genotypes, which will be added to the GeneLab dataset.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will lead to a better understanding of the role of Putrescine and/or its derived signaling products in mitigating the stress associated with growth under microgravity environments such as those encountered during spaceflight and on the International Space Station. It may also allow the identification of novel genetic engineering strategies aimed at improving plant adaptation to the microgravity environment, with potential impact on our ability to improve the genome of other plant species / crops for utilization in bioregenerative life-support systems during spaceflight. A better understanding of plant responses to stress may also have important applications in agriculture for crop production on marginal lands. New large-scale RNAseq expression data associated with growth under microgravity for previously untested Arabidopsis genotypes will also be generated during this project and submitted to the GeneLab Data System. This project will allow us to introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, and also generate novel instructional materials allowing effective teaching of the difficult concepts of gravity perception by plants and their importance for agricultural production on Earth and during spaceflight. Such materials should be usable in K-12, undergraduate, and graduate education, as well as general-public outreach.
Task Progress:	The main objective of Advanced Plant Experiment (APEX-08) is to evaluate the impact of genetic alterations of the putrescine pool in Arabidopsis thaliana on its seedlings' ability to grow, develop, and respond transcriptionally to the microgravity conditions of the International Space Station (ISS) relative to ground control. To this end, we developed and characterized Arabidopsis thaliana mutant and transgenic lines with alterations in putrescine metabolism including: (1) wild type Col-0; (2) adc1-1 knockout mutant affected in its ability to synthesize Put; (3) Cuao3-1 and cuao3-100cdr7 allelic mutants affected in their ability to catabolize putrescine; (4) transgenic plants over-expressing the Poncirus trifoliata ADC (PtADC) gene involved in putrescine synthesis; and (5) transgenic cuao3 plants over-expressing the AtCuAO3 transgene. We then used these genotypes to evaluate the effect of alterations in the pool of putrescine and/or its catabolites on Arabidopsis thaliana seedling responses to the stress encountered during spaceflight. Results indicate that seedlings of all genotypes grown under microgravity conditions on the ISS carry longer leaf petioles than those grown in a ground control at NASA Kennedy Space Center (KSC). Furthermore, plants over-expressing the PtADC gene display simplified transcriptional response profiles to spaceflight relative to ground control, with weaker oxidative stress response; whereas those with a knockout mutation in the CuAO3 gene develop more complex response profiles to spaceflight. Furthermore, genes associated with epigenetic regulatory processes such as DNA and histone modifications are differentially regulated between spaceflight and ground control conditions in a genotype- and organ-specific manner. Together, these results suggest increased accumulation of putrescine, and/or putrescine-derived compounds, leads to enhanced tolerance to spaceflight. Considering this conclusion, we also set up to investigate the ability of plants with the same alterations in the PA metabolic pathway to grow and develop on lunar regolith simulant (LHS-1), a substrate previously suggested to be stressful for plants. The results of preliminary seedling-viability experiments suggest plants over-expressing the PtADC gene also display increased tolerance to lunar regolith simulants. Our results suggest a novel genetic engineering strategy to engineer plants with improved tolerance to spaceflight and/or lunar regolith simulant, with important implications for space exploration programs.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Articles in Peer-reviewed Journals	Su S-H, Levine HG, Masson PH. "Brachypodium distachyon seedlings display accession-specific morphological and transcriptomic responses to the microgravity environment of the International Space Station." Life. 2023 Feb 23;13(3):626. https://doi.org/10.3390/life13030626 ; PMID: 36983782; PMCID: PMC10058394 , Feb-2023
Awards	Su S-H. "University of Wisconsin-Madison Award for Mentoring Undergraduates in Research, Scholarly and Creative Activities. " May-2023